

ShopEZ – Brainstorming & Ideation

Date: 18 June 2026

Project Name: ShopEZ – E-Commerce Shopping Platform

Developer: Individual Project

Maximum Marks: 4 Marks

STEP 1: PROBLEM STATEMENT SELECTION

Selected Problem Statement:

Many customers face difficulties while shopping online due to complex interfaces, limited product information, and inefficient checkout processes. Small business owners also face challenges in managing products, inventory, and customer orders effectively.

ShopEZ addresses these issues by providing a modern e-commerce platform that offers easy product discovery, secure transactions, responsive design, and efficient administration tools for business management.

STEP 2: BRAINSTORM IDEAS

Theme: User Experience

Ideas Generated: • Responsive and mobile-friendly design • Easy navigation and clean interface • Advanced product search functionality • Product filtering and sorting • Dark mode support • Quick checkout process

Theme: Product Management

Ideas Generated: • Product categorization • Product image gallery • Inventory tracking • Product reviews and ratings • Featured product listings • Promotional offers and discounts

Theme: Shopping & Orders

Ideas Generated: • Shopping cart management • Secure checkout process • Order tracking • Order history • Wishlist functionality • Email order confirmations

Theme: Administration

Ideas Generated: • Product CRUD operations • User management dashboard • Order monitoring • Sales reporting • Inventory management • Analytics dashboard

Theme: Security

Ideas Generated: • JWT Authentication • Password encryption using bcrypt.js • Role-based authorization • Protected routes • Secure API access • Input validation

Theme: Analytics

Ideas Generated: • Sales performance charts • Revenue reports • Customer activity tracking • Product popularity metrics • Order trend visualization • Category-wise sales analysis

STEP 3: IDEA PRIORITIZATION MATRIX

Idea	Importance	Feasibility	Priority
User Authentication	High	High	Do First
Product Catalog	High	High	Do First
Shopping Cart	High	High	Do First
Checkout System	High	High	Do First
Product Search & Filters	High	Medium	Do Next
Order Tracking	High	Medium	Do Next
Admin Dashboard	Medium	High	Do Later
Analytics Dashboard	Medium	Medium	Do Later
Product Recommendations	Medium	Low	Future Enhancement
AI-Based Suggestions	Low	Low	Future Enhancement

CONCLUSION

The brainstorming process helped identify the core features required for the ShopEZ platform. User authentication, product browsing, shopping cart management, and checkout functionality were prioritized as the minimum viable product (MVP). Additional analytics and recommendation features are planned as future enhancements to improve user experience and business insights.